

Quantum gates and algorithms in qg units: Hamming-weight conservation and readout classes

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Abstract

We write fifteen standard quantum gates and fourteen standard quantum algorithms in qg units [1]. An n -qubit state is described by its qg values, $qg_P = \langle P \rangle$ for every Pauli string P , and a gate U acts as $qg'_P = qg_{U^\dagger P U}$. With this one rule each gate becomes a table of qg transformations and each algorithm a readout identity: which qg values hold the answer. The formulation is the Pauli (Heisenberg) representation and adds no new physics. It gives three practical classifications in the units used by the rest of the qang project: which gates conserve Hamming weight, and therefore admit the qg symmetry witness and filter; which algorithms are read out from a few qg values and which need the full outcome histogram; and where reading qg values is strictly worse than the standard readout. Three readout identities are, as far as we know, not stated in this form elsewhere: the per-qubit polar bias after Grover iterations, Simon's secret as the unique non-trivial Z-parity equal to one, and quantum counting as a single qg_X . Every identity is checked numerically and pinned by tests in the open-source library `qang` (version 0.5.1, `pip install qang`), module `qang.formulation`.

1 Why write gates and algorithms in qg units

The qang project describes a qubit by measurement-level quantities, the polar bias $qg_Z = \langle \sigma_z \rangle$ and its companions [1], and builds diagnostics on them: a register-mean witness and a shot-level filter for circuits that conserve a Hamming weight [2], and decoders that read relaxation from the final readout [3]. Those tools apply to some circuits and not to others. The question of this note is the bookkeeping one: for the gates and algorithms that people actually use, where do the qg tools apply, and what does a qg readout give? We answer it with the Pauli representation, which is standard [4, 5], stated in qg units and checked numerically line by line.

2 Framework

State. For one qubit, $\rho = \frac{1}{2}(I + qg_X X + qg_Y Y + qg_Z Z)$; for n qubits,

$$\rho = \frac{1}{2^n} \sum_{P \in \{I, X, Y, Z\}^{\otimes n}} qg_P P, \quad qg_P = \text{Tr}(\rho P), \quad (1)$$

and the purity is $\text{Tr}\rho^2 = 2^{-n} \sum_P qg_P^2$. Strings with one non-identity letter give the local qg values; the others are correlations, such as qg_{ZZ} .

Measurement. A Z-basis measurement gives, from the same shots, every diagonal value $qg_{Z_S} = \sum_y p(y) (-1)^{S \cdot y}$ for every subset S of qubits; for one qubit $P(0) = (1 + qg_Z)/2$. Values with X or Y need a basis rotation first.

Gates. A gate U maps qg values as

$$qg'_P = qg_{U^\dagger P U}. \quad (2)$$

For a Clifford gate, $U^\dagger P U$ is a signed Pauli string and the gate permutes qg values with signs; for a non-Clifford gate it is a combination, and the gate mixes them. Convention: the leftmost letter of a string is qubit 0, which is the control of CX and CZ (and the controls of Toffoli and Fredkin).

Weight conservation. If U maps each Hamming-weight sector into itself, the register mean $\overline{qg_Z} = 1 - 2N/n$ stays fixed, and the qg witness and filter apply.

3 Fifteen gates

Table 1: Action of each gate on qg values, Eq. (2); $c = \cos \theta$, $s = \sin \theta$. Values not listed are unchanged.

gate	action on qg values	reading
1. Pauli X	$(qg_X, qg_Y, qg_Z) \rightarrow (qg_X, -qg_Y, -qg_Z)$	flips the polar bias
2. Pauli Y	$(qg_X, qg_Y, qg_Z) \rightarrow (-qg_X, qg_Y, -qg_Z)$	flips qg_Z and qg_X
3. Pauli Z	$(qg_X, qg_Y, qg_Z) \rightarrow (-qg_X, -qg_Y, qg_Z)$	invisible in the Z basis
4. Hadamard	$(qg_X, qg_Y, qg_Z) \rightarrow (qg_Z, -qg_Y, qg_X)$	swaps polar bias and qg_X
5. Phase S $P(\varphi)$	$(qg_X, qg_Y) \rightarrow (-qg_Y, qg_X)$ $qg'_X = \cos \varphi qg_X - \sin \varphi qg_Y$, $qg'_Y = \sin \varphi qg_X + \cos \varphi qg_Y$	quarter turn on the equator same as $R_z(\varphi)$
6. T ($\pi/8$)	$qg'_X = (qg_X - qg_Y)/\sqrt{2}$, $qg'_Y = (qg_X + qg_Y)/\sqrt{2}$	first gate that mixes: irrational qg values (“magic”)
7. $R_x(\theta)$	$qg'_Z = c qg_Z + s qg_Y$, $qg'_Y = c qg_Y - s qg_Z$	rotation in the YZ plane
8. $R_y(\theta)$	$qg'_Z = c qg_Z - s qg_X$, $qg'_X = c qg_X + s qg_Z$	from $ 0\rangle$: $qg_Z = \cos \theta$, the qang identity; the RQang gate is $R_y(\arccos qg_Z)$
9. $R_z(\theta)$	$qg'_X = c qg_X - s qg_Y$, $qg'_Y = s qg_X + c qg_Y$	invisible in the Z basis
10. CNOT	$qg'_{XI} = qg_{XX}, qg'_{IZ} = qg_{ZZ}$, $qg'_{IY} = qg_{ZY}, qg'_{YI} = qg_{YX}$	turns a correlation into a local value: the target’s qg_Z becomes the parity qg_{ZZ} (syndrome extraction)
11. CZ	$qg'_{XI} = qg_{XZ}, qg'_{IX} = qg_{ZX}$, $qg'_{YI} = qg_{YZ}, qg'_{IY} = qg_{ZY}$	invisible in the Z basis; conserves weight
12. SWAP	$qg'_{ab} = qg_{ba}$	relabels; conserves weight
13. iSWAP	$qg'_{ZI} = qg_{IZ}, qg'_{XI} = -qg_{ZY}$, $qg'_{YI} = qg_{ZX}, qg'_{IX} = -qg_{YZ}$, $qg'_{IY} = qg_{XZ}$	swap with phase; conserves weight
14. Toffoli	$qg'_{IIZ} =$ $\frac{1}{2}(qg_{IIZ} + qg_{IZZ} + qg_{ZIZ} - qg_{ZZZ})$	mixes inside the diagonal sector
15. Fredkin	$qg'_{IZI} =$ $\frac{1}{2}(qg_{IZI} + qg_{ZZI} + qg_{IIZ} - qg_{ZIZ})$	controlled swap; conserves weight

Three observations organize the table.

- **Clifford or not.** X, Y, Z, H, S, CNOT, CZ, SWAP and iSWAP permute qg values with signs. T, generic rotations, Toffoli and Fredkin mix them: one string opens into two (T) or four (Toffoli, Fredkin) per application. That growth is what makes non-Clifford circuits hard to simulate [4].
- **Z-basis blindness.** Z, S, T, R_z , $P(\varphi)$ and CZ leave every qg_Z fixed; their effect appears only after a gate that moves the equator to the poles. This is, gate by gate, the Z-basis blind spot of the qg metrics [1].
- **Weight conservation.** Among multi-qubit gates, CZ, SWAP, iSWAP and Fredkin conserve Hamming weight; CNOT and Toffoli do not. A circuit built only from weight-conserving gates and R_z rotations has zero sector exposure, and in our simulations the qg filter then removes almost all relaxation error; basis changes around two-qubit gates let about half of it through [2].

4 Fourteen algorithms

For each: the readout identity, what reading it in qg gives, and what it does not.

1. Deutsch–Jozsa [6]. With f as a phase oracle, $P(0^n) = 2^{-n} \sum_S qg_{Z_S}$: the all-zero probability is the mean of all Z-parities. In qg: f is constant if and only if all n local qg_Z equal +1; one value below 1 already means balanced. No correlation is needed.

2. Bernstein–Vazirani [7]. The output is the product state $|s\rangle$, so $qg_Z^{(i)} = (-1)^{s_i}$: each secret bit is the sign of its qubit’s polar bias. Under noise each bit is decided separately, with its own interval.

3. Simon [8]. The output is uniform over $\{y : y \cdot s = 0\}$, so $qg_{Z_S} = 1$ for $S \in \{0, s\}$ and 0 otherwise: the secret is the unique non-trivial Z-parity equal to one. This describes the answer but does not find it efficiently, since there are 2^n parities; the efficient route remains a linear system from $O(n)$ samples.

4. Shor [9]. The quantum part estimates the phase of an order r . If r is a power of two, the local values show it: for $r = 4$ with six counting qubits, $qg_Z = (0, 0, 1, 1, 1, 1)$. For $r = 6$ the local qg_Z lie between 0 and 0.33 and do not determine r ; the information is in the correlations, and the full bit string plus continued fractions is needed. This is a negative result for local readout.

5. Grover [10]. With one marked item x^* among $N = 2^n$ and success probability P after k iterations,

$$qg_Z^{(i)} = (-1)^{x_i^*} \frac{NP - 1}{N - 1}. \quad (3)$$

The marked item is the sign pattern of the local polar biases, and their common magnitude measures success. Reading it this way costs more shots than the histogram, about $1/P^2$ against $1/P$, as we found in a noisy simulation; the identity is a diagnostic, not a better readout.

6. Phase estimation [11] and 11. phase kickback. A control in $|+\rangle$ acting on an eigenstate of U with eigenvalue $e^{i\varphi}$ ends with

$$qg_X + i qg_Y = e^{i\varphi}, \quad qg_Z \text{ unchanged.} \quad (4)$$

Kickback writes the phase into the control’s equatorial angle. Iterative phase estimation reads qg_X and qg_Y for $2^k \varphi$, one bit at a time; it is the natural qg form of phase estimation, and the same mechanism drives Deutsch–Jozsa and Bernstein–Vazirani.

7. HHL [12]. The output $|x\rangle \propto A^{-1}|b\rangle$ cannot be read out in full; what HHL delivers is expectation values $\langle x|M|x\rangle$, that is qg values of the solution register, and its success probability is the ancilla’s polar bias, $P(1) = (1 - qg_Z^{\text{anc}})/2$. The qg form states plainly what HHL produces: qg values, not the vector [13]. We check this readout for a 2×2 matrix by exact linear algebra, not with a circuit.

8. Quantum Fourier transform. The QFT of a basis state $|x\rangle$ is a product of equatorial qubits: qubit l has $qg_Z = 0$ and $qg_X + i qg_Y = e^{2\pi i x/2^l}$. The QFT moves the information from polar biases (the bits of x) to equatorial angles, which the Z basis does not see; hence it is always followed by a basis change before measurement.

9. VQE [14]. The energy is linear in qg values, $E = \sum_P c_P qg_P$, grouped by measurement basis. This is the main use of the qg tools so far: witness, filter, symmetry checks and pole-damped optimization [1, 2].

10. QAOA [15]. For MaxCut, $C = \sum_{(i,j)} w_{ij}(1 - qg_{Z_i Z_j})/2$: the whole cost comes from ZZ correlations in the same shots. With a weight-conserving XY mixer the witness and filter apply [2].

12. Quantum counting [16]. A Hadamard test of the Grover operator G on the uniform state $|s\rangle$ leaves the control with

$$qg_X = \langle s|G|s\rangle = 1 - \frac{2M}{N} \implies M = \frac{N(1 - qg_X)}{2}. \quad (5)$$

Counting solutions is measuring one qg_X : with $N = 16$ and three marked items, $qg_X = 0.625$ and $M = 3$. The precision follows single-qubit statistics; full counting by phase estimation gains precision with more counting qubits.

13. Quantum walks [17]. A continuous-time single walker with hopping $(XX + YY)/2$ conserves the number of excitations, so the register mean stays at $\overline{qg_Z} = 1 - 2/n$ at all times (checked for $n = 5$): single-walker walks live in the weight-one sector, where the witness and filter apply as in Heisenberg dynamics. Discrete coined walks do not conserve weight.

14. Machine learning: Q-SVM [18], Q-PCA [19]. The fidelity kernel is an all-zero probability, $k(x, x') = P(0^n) = 2^{-n} \sum_S qg_{Z_S}$, and for product angle encodings it reduces to $\prod_i (1 + \vec{q}_i \cdot \vec{q}_i')/2$, the qang transition probability. In Q-PCA the purity and spectrum of ρ depend on squared qg values, $\text{Tr}\rho^2 = 2^{-n} \sum_P qg_P^2$.

5 Readout classes

Table 2: Which qg values suffice to read out each algorithm.

sufficient readout	algorithms
local qg_Z	Deutsch–Jozsa, Bernstein–Vazirani
one ancilla’s qg_X, qg_Y	phase kickback, iterative phase estimation, quantum counting (Hadamard test), HHL success
few-qubit correlations	QAOA (ZZ), VQE (Hamiltonian strings)
all parities or the histogram	Simon, Shor with general order, Grover (local readout exists, at $\sim 1/P^2$ shots), fidelity kernels

6 What is new and what is not

The Pauli representation of states, the action of gates by conjugation, and the stabilizer formalism are known [4, 5], and so are the algorithms [6–12, 14–19]. The contribution is organizational and practical: one language, the qg units of the qang project, for three questions. (i) Which gates conserve weight and admit the qg witness and filter. (ii) Which algorithms are read out from a few qg values and which need the histogram. (iii) Where qg readout is worse (Grover, Shor with general order, Simon). The readout forms in Eqs. (3) and (5), and Simon’s secret as the unique unit parity, are elementary but, to our knowledge, not stated in this form elsewhere. None of this is a quantum advantage.

7 Limitations

(i) Ideal, noiseless states throughout; noise enters only through the cited qg studies. (ii) The identities are checked on small instances ($n \leq 6$). (iii) The HHL readout is checked by exact linear algebra for a 2×2 matrix, not with a circuit. (iv) Shot costs are quoted from a separate simulation (Grover) or are the standard single-qubit statistics.

8 Code availability

The library `qang` (MIT license) is on PyPI (`pip install qang`, version 0.5.1) and at <https://github.com/Viny2030/qang>. The module `qang.formulation` gives the gate matrices, the rule of Eq. (2) (`conjugate`, `gate_table`, `apply_gate`), qg values of a state (`qg_values`) and one readout function per algorithm; `tests/test_formulation.py` pins every identity, and `notebooks/qang_inicio_formulacion.ipynb` runs them in Colab. For example:

```
from qang import formulation as F
bell = F.apply_gate(F.apply_gate(F.qg_values([1, 0, 0, 0]),
                                     F.kron(F.H, F.I2)), F.CX)
# qg_XX = 1, qg_YY = -1, qg_ZZ = 1, all local values 0
```

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